Handwritten Digit Classification on MNIST Subset

CS5487 Machine Learning – Course Project Report

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Abstract

This report presents an empirical study of multiple classification methods for handwritten digit recognition on a 4,000-sample MNIST subset. We evaluate eleven classifier configurations across two predefined train/test splits, comparing k-Nearest Neighbors (k-NN), Linear Discriminant Analysis (LDA), Quadratic Discriminant Analysis (QDA), Support Vector Machines (SVM) with linear and RBF kernels, Logistic Regression, and data-augmented variants. Feature preprocessing techniques including scaling, Principal Component Analysis (PCA), and geometric data augmentation (rotation, translation, scaling) are investigated. Our best model, SVM with RBF kernel combined with PCA to 50 components and mild data augmentation ($\times 3$), achieves **96.55%** mean accuracy on the MNIST test set, outperforming the 1-NN baseline of 91.60%. On a held-out challenge dataset of 150 professor-written digits, the augmented model achieves **83.33%**, a 15.03 percentage point improvement over the baseline. Detailed error analysis reveals that 1.5% of samples are "hard" (misclassified by all models), while 14.1% are "medium" where model capability differences matter.

1 Introduction

Handwritten digit recognition is a classic problem in machine learning and computer vision, serving as a benchmark for evaluating classification algorithms. The MNIST dataset [1] has been the standard testbed for decades. In this project, we work with a curated 4,000-sample subset of MNIST containing 400 images per digit class (0–9), where each image is a 28×28 grayscale pixel map vectorized into 784-dimensional feature vectors with pixel values in $[0, 255]$.

The project evaluates several classification methods covered in the CS5487 course, including:

- k-Nearest Neighbors (k-NN)
- Linear Discriminant Analysis (LDA)
- Quadratic Discriminant Analysis (QDA)
- Support Vector Machines (SVM) with linear and RBF kernels
- Logistic Regression

Additionally, we investigate the effect of feature preprocessing (normalization, PCA dimensionality reduction) and geometric data augmentation. We evaluate all trained models on a challenge dataset consisting of the professor's own handwritten digits, without any retraining or parameter tuning on the challenge data.

2 Methodology

2.1 Dataset and Evaluation Protocol

The dataset is stored in a MATLAB `.mat` file with the following fields:

- `digits_vec`: 784×4000 matrix, each column is a flattened image.

- `digits_labels`: 1×4000 vector of class labels $y \in \{0, \ldots, 9\}$.
- `trainset`: 2×2000 matrix, two predefined training index sets.
- `testset`: 2×2000 matrix, two corresponding test index sets.

The dataset defines two fixed train/test splits (Trial 1 and Trial 2). In each trial, 2,000 samples (50%) are used for training and 2,000 for testing. Importantly, the same writer does not appear in both training and test sets within a trial, ensuring writer-independent generalization. All reported accuracies are averaged over both trials with standard deviation.

2.2 Classifiers

2.2.1 k-Nearest Neighbors (k-NN)

k-NN is a non-parametric baseline classifier. We use Euclidean distance and test $k \in \{1, 3, 5\}$. No training is required; prediction is by majority vote among the k closest training examples.

2.2.2 Linear Discriminant Analysis (LDA)

LDA assumes each class follows a Gaussian distribution with a shared covariance matrix. The decision boundary is linear. We use the standard eigenvalue decomposition solver.

2.2.3 Quadratic Discriminant Analysis (QDA)

QDA relaxes the shared covariance assumption, allowing each class its own covariance matrix. This yields quadratic decision boundaries. Since QDA requires estimating $10 \times 784 \times 784$ covariance parameters from only 2,000 training samples, we apply PCA to reduce dimensionality to 50 before fitting QDA. PCA is fit on the training data only.

2.2.4 Support Vector Machines (SVM)

SVM finds the maximum-margin hyperplane in a potentially high-dimensional feature space. We evaluate:

- **SVM-Linear**: Linear kernel, $C = 1.0$.
- **SVM-RBF**: Radial Basis Function kernel with $\gamma = \text{scale}$, $C = 1.0$.
- **SVM-RBF + PCA50**: RBF kernel after PCA to 50 components.

Multi-class SVM is implemented via a one-vs-all (OvA) wrapper: 10 binary classifiers are trained, one per digit. At test time, the classifier with the highest decision function value (furthest from the margin) is selected.

2.2.5 Logistic Regression

We use L_2-regularized multinomial logistic regression with $C = 1.0$ and a maximum of 1,000 iterations, also using the one-vs-all strategy. The features are standardized to zero mean and unit variance.

2.2.6 Data Augmentation

To improve generalization across different handwriting styles, we apply controlled geometric transformations to training images before normalization and PCA:

- **Rotation**: random angle in $[-5, +5]$ (mild), $[-10, +10]$ (medium), or $[-15, +15]$ (strong)
- **Translation**: random shift of ±1px (mild), ±2px (medium), or ±3px (strong)
- **Scaling**: random scale factor 0.95–1.05 (mild), 0.90–1.10 (medium), or 0.85–1.15 (strong)
- **Strong mode**: additional Gaussian noise and brightness adjustment

Each training sample is augmented with n synthetic copies, yielding $(1 + n) \times 2{,}000$ training samples. Augmentation is applied *before* normalization and PCA, and the same augmentation pipeline is used for both trials.

2.3 Feature Preprocessing

Table 1: Preprocessing strategies per classifier.

Classifier	**Normalization**	**PCA**	**Augmentation**
1-NN, 3-NN, 5-NN	None	None	None
LDA	None	None	None
QDA + PCA50	None	50 components	None
SVM-Linear	Scale to $[0, 1]$	None	None
SVM-RBF	Scale to $[0, 1]$	None	None
SVM-RBF + PCA50	Scale to $[0, 1]$	50 components	None
Logistic Regression	Standard ($\mu = 0, \sigma = 1$)	None	None
SVM-RBF + PCA35 + Aug	Scale to $[0, 1]$	35 components	Mild $\times 3$
SVM-RBF + PCA50 + Aug	Scale to $[0, 1]$	50 components	Mild $\times 3$

We experiment with three normalization schemes:

- **None**: Raw pixel values in $[0, 255]$.
- **Scale255**: Divide by 255 to map to $[0, 1]$.
- **Standard**: Per-feature standardization using training set statistics.

PCA is fit exclusively on the training data and applied to both training and test data to avoid data leakage.

3 Experimental Setup

All experiments were implemented in Python using scikit-learn [2]. The code is organized into modular components:

- `data_loader.py`: Data loading, train/test splitting, normalization.
- `classifiers.py`: Classifier factory, one-vs-all wrapper, PCA transform.
- `augmentation.py`: Geometric data augmentation (rotation, translation, scaling).
- `experiments.py`: Experiment runner executing both trials.
- Analysis scripts: Error analysis, sample-level error analysis, class distribution visualization, and challenge evaluation.

Evaluation Metric. The primary evaluation metric is **classification accuracy** (number of correct predictions divided by total number of samples). All reported accuracies are averaged over the two predefined train/test splits with standard deviation.

Cross-Validation for Parameter Selection. To avoid tuning parameters on the test set, we use 3-fold stratified cross-validation on the *training data only* to select the optimal data augmentation strategy and SVM hyperparameters. The challenge dataset is completely held out and used only for final reporting.

Challenge Dataset. A held-out challenge dataset of 150 professor-written digits is used to evaluate cross-writer generalization. Models are trained on the MNIST training splits and evaluated on the challenge data *without retraining or parameter tuning*.

Results are managed in a unified directory structure: `results/<git-hash>/<timestamp>/`, with subdirectories for figures and logs, ensuring reproducibility.

4 Experimental Results

4.1 MNIST Subset Classification Results

Table 2 reports the accuracy of each method on the two predefined trials.

Table 2: Classification accuracy on the MNIST 4,000-sample subset. Best result in **bold**.

Method	Trial 1	Trial 2	Mean ± Std
1-NN Baseline	0.9135	0.9185	0.9160 ± 0.0035
3-NN	0.9180	0.9060	0.9120 ± 0.0060
5-NN	0.9170	0.9095	0.9133 ± 0.0038
LDA	0.7845	0.7795	0.7820 ± 0.0025
QDA + PCA50	0.9330	0.9430	0.9380 ± 0.0050
SVM-Linear	0.8600	0.8580	0.8590 ± 0.0010
SVM-RBF	0.9395	0.9335	0.9365 ± 0.0030
SVM-RBF + PCA50	0.9510	0.9470	0.9490 ± 0.0020
Logistic Regression	0.8680	0.8640	0.8660 ± 0.0020
SVM-RBF + PCA35 + Aug	0.9650	0.9625	0.9637 ± 0.0012
SVM-RBF + PCA50 + Aug	**0.9675**	**0.9635**	**0.9655 ± 0.0020**

Key observations:

1. **SVM-RBF + PCA50 + Aug** achieves the highest mean accuracy (96.55%), outperforming the 1-NN baseline by 4.95 percentage points and the non-augmented SVM-RBF + PCA50 by 1.65 percentage points.
2. Data augmentation provides substantial gains: both augmented SVM variants exceed 96%, while their non-augmented counterparts plateau around 94–95%. Mild geometric transformations introduce valuable training diversity without distorting digit structure.
3. PCA helps SVM-RBF significantly: accuracy improves from 93.65% (no PCA) to 94.90% (PCA50). PCA removes noisy high-frequency pixel variations, allowing the RBF kernel to focus on discriminative low-dimensional structure.
4. QDA also benefits from PCA (93.80%), as the full 784-dimensional covariance estimation is ill-posed with only 2,000 training samples.
5. LDA performs poorly (78.20%), suggesting the shared-covariance Gaussian assumption is too restrictive for this data.
6. SVM-Linear (85.90%) and Logistic Regression (86.60%) are competitive but trail nonlinear methods, indicating that digit classes are not linearly separable in pixel space.
7. k-NN with $k > 1$ does not improve over 1-NN; in fact, 3-NN and 5-NN slightly underperform, suggesting that the nearest neighbor alone is the most reliable predictor in this feature space.

4.2 Challenge Dataset Evaluation

The challenge dataset contains 150 handwritten digits written by the course instructor. Models trained on the MNIST training split are evaluated directly on this data without retraining or parameter tuning. Table 3 reports the results.

Table 3: Challenge dataset accuracy (150 samples). Reference 1-NN baseline: 68.33%.

Method	Trial 1	Trial 2	Mean ± Std
1-NN Baseline	0.6600	0.7067	0.6833 ± 0.0233
3-NN	0.6667	0.6400	0.6533 ± 0.0133
5-NN	0.6333	0.6467	0.6400 ± 0.0067
LDA	0.4533	0.4200	0.4367 ± 0.0167
QDA + PCA50	0.6933	0.7267	0.7100 ± 0.0167
SVM-Linear	0.6133	0.5533	0.5833 ± 0.0300
SVM-RBF	0.6867	0.7000	0.6933 ± 0.0067
SVM-RBF + PCA50	0.7000	0.7333	0.7167 ± 0.0167
Logistic Regression	0.5800	0.6000	0.5900 ± 0.0100
SVM-RBF + PCA35 + Aug	0.8200	0.8467	0.8333 ± 0.0133
SVM-RBF + PCA50 + Aug	**0.8200**	**0.8467**	**0.8333 ± 0.0133**

Key observations:

1. **SVM-RBF + PCA50 + Aug** achieves the best result (83.33%), improving over the 1-NN baseline by 15.03 percentage points and over the non-augmented SVM-RBF + PCA50 by 11.67 percentage points.
2. Data augmentation dramatically improves challenge generalization. While non-augmented models drop 20–25 percentage points from MNIST to challenge, the augmented model drops only 13.2 points (96.55% to 83.33%), demonstrating that geometric transformations help the model learn writer-invariant features.
3. All non-augmented methods suffer a significant accuracy drop on the challenge data, indicating a substantial domain shift between student-written MNIST digits and professor-written challenge digits.
4. QDA + PCA50 (72.33%) is the best non-augmented method, confirming that PCA-based dimensionality reduction yields robust classifiers.
5. k-NN degrades significantly on the challenge set, suggesting that the local neighborhood structure in pixel space does not transfer well across different writing styles.

4.3 Parameter Sensitivity Analysis

To understand how hyperparameters affect performance, we conduct two controlled sensitivity studies using 3-fold stratified cross-validation on the training data only.

4.3.1 Data Augmentation Intensity

Table **??** compares different augmentation configurations via cross-validation. The base model is PCA(35) + SVM-RBF($C = 4$, $\gamma =$ scale).

Table 4: 3-fold cross-validation scores for data augmentation strategies.

Configuration	CV Mean	CV Std
Original (no augmentation)	94.45%	0.56%
Mild augmentation ×3	**98.73%**	**0.27%**
Medium augmentation ×3	95.69%	0.50%
Strong augmentation ×3	60.47%	0.18%
Medium augmentation ×5	96.79%	0.21%
Medium augmentation ×7	97.61%	0.13%

Key findings:

- **Mild augmentation is optimal.** The ± 5 rotation, ± 1px translation, and 0.95–1.05 scaling preserve digit structure while providing regularization benefits.
- **Strong augmentation hurts.** Excessive distortion (noise, brightness changes, large rotations) degrades CV accuracy to 60.47%, confirming that destructive augmentation destroys useful signal.
- Medium augmentation ($\times 3$ and $\times 5$) improves over no augmentation but underperforms mild augmentation, suggesting that even moderate transformations introduce too much variance for this dataset.

4.3.2 SVM Hyperparameter and PCA Component Selection

Table **??** shows the result of a grid search over PCA components and SVM regularization parameter C, evaluated via 3-fold cross-validation on the training data.

Table 5: GridSearchCV results (3-fold CV) for PCA components $\times$ SVM C. Best in **bold**.

PCA	$C = 3$	$C = 4$	$C = 5$	$C = 6$	$C = 8$
20	94.85%	95.05%	95.00%	94.95%	94.90%
25	95.10%	95.20%	95.15%	95.10%	95.05%
30	95.25%	95.30%	95.25%	95.20%	95.15%
35	95.30%	**95.35%**	95.30%	95.25%	95.20%
40	95.25%	**95.35%**	95.30%	95.25%	95.20%

Key findings:

- **Optimal region: PCA 35–40, $C = 4$.** CV accuracy peaks at 95.35%, with little sensitivity around this region.
- Performance degrades slightly with fewer components (20–25) and with overly large C (≥ 8), suggesting mild overfitting when too many principal components or too little regularization is used.
- The chosen configuration (PCA=35, $C = 4$) generalizes well to both test trials, confirming that cross-validation on training data alone is sufficient for parameter selection.

4.4 Error Analysis

4.4.1 Confusion Patterns

We analyze the confusion matrices of the best-performing model (SVM-RBF + PCA50) across both trials. The top confused digit pairs are:

- $5 \to 3$: 11 errors (2.75% of digit 5 samples)
- $4 \to 9$: 9 errors (2.25%)
- $9 \to 7$: 9 errors (2.25%)
- $5 \to 6$: 8 errors (2.00%)
- $9 \to 4$: 8 errors (2.00%)

The most frequent confusions involve visually similar digits: 5 vs. 3 (both have a top curve and bottom loop), 4 vs. 9 (share an upper loop), and 9 vs. 7 (both have a top horizontal stroke).

Per-digit accuracy for SVM-RBF + PCA50 is high across all classes, with the lowest accuracy on digit 3 (93.00%) and digit 9 (92.75%). Digit 1 is the easiest (98.75%), followed by digit 0 (97.75%).

4.4.2 Sample-Level Error Analysis

To understand whether errors are sample-specific (hard samples) or model-specific, we compare predictions across five models (1-NN, 3-NN, SVM-RBF, SVM-RBF+PCA50, QDA+PCA50) on all 4,000 test samples:

- **Hard samples** (all 5 models wrong): 60 samples (1.5%). These are likely genuinely ambiguous, poorly written, or potentially mislabeled digits. Model improvements will not help here.
- **Medium samples** (some models wrong): 563 samples (14.1%). These indicate model capability differences—better models can improve on these.
- **Easy samples** (all models correct): 3,377 samples (84.4%).

Among the medium samples, models agree on the wrong prediction for 461 samples but disagree for 102 samples, suggesting that for most errors, even the wrong predictions are structurally similar.

4.5 Class Distribution

The MNIST subset is perfectly balanced with 400 samples per digit class (10.0% each). Both predefined train/test splits maintain this balance exactly (200 training / 200 test per class per trial). This eliminates class imbalance as a confounding factor in our experiments.

5 Discussion

Effect of data augmentation. Mild geometric augmentation (rotation ± 5, translation ± 1px, scale 0.95–1.05, $\times 3$ copies per sample) is the single most impactful improvement in this study. It raises MNIST test accuracy from 94.90% to 96.55% and challenge accuracy from 71.67% to 83.33%. The augmentation introduces controlled diversity into the training set, helping the model learn features invariant to small pose variations. Importantly, stronger augmentation with noise and brightness changes actually degrades performance, confirming that excessive distortion destroys useful signal. The optimal augmentation is light enough to preserve digit structure while providing regularization benefits.

Effect of PCA. PCA consistently improves non-linear methods (SVM-RBF and QDA). For the augmented SVM, reducing from 50 to 35 components yields nearly identical performance, suggesting that 35 components capture sufficient discriminative information. The 35–50 components explain approximately 80–83% of the variance. By discarding high-frequency noise and redundancy in the 784-dimensional pixel space, PCA allows the classifiers to operate in a more structured, lower-dimensional manifold where class separation is cleaner.

Effect of normalization. Scaling pixel values to $[0, 1]$ is essential for SVMs with RBF kernels, as the kernel bandwidth γ is computed relative to feature scales. Without scaling, pixel value magnitudes dominate the distance computation. Standardization (zero mean, unit variance) works well for Logistic Regression but is unnecessary for k-NN and tree-based methods.

Linear vs. non-linear. The large gap between linear methods (LDA: 78.20%, SVM-Linear: 85.90%, Logistic Regression: 86.60%) and non-linear methods (SVM-RBF: 93.65–96.55%) confirms that digit classes have complex, non-linear decision boundaries in pixel space. This justifies the use of kernel methods and quadratic discriminant functions.

Domain shift. The challenge results highlight a critical issue in handwritten digit recognition: models trained on one writer population do not generalize perfectly to a different writer. Non-augmented models suffer a 20–25 percentage point drop in accuracy. However, data augmentation effectively closes this gap: the augmented model drops only 13.2 points (96.55% to 83.33%), demonstrating that geometric transformations help the model learn writer-invariant features. This suggests that data augmentation is a practical and effective technique for improving cross-writer generalization without requiring additional training data or domain adaptation methods.

6 Conclusion

We conducted a comprehensive empirical evaluation of eleven classifier configurations for handwritten digit classification, including data-augmented variants. All parameters were selected using 3-fold cross-validation on the training data only, with the challenge dataset held out for final evaluation. Our main findings are:

1. **SVM with RBF kernel, PCA to 50 components, and mild data augmentation ($\times$3)** achieves the best performance on both the MNIST subset (96.55%) and the challenge dataset (83.33%).

2. **Data augmentation is the single most impactful improvement**: it raises challenge accuracy by 11.67 percentage points over the non-augmented model, effectively closing the generalization gap between student-written and professor-written digits. Cross-validation confirms mild augmentation (CV 98.73%) outperforms both no augmentation (94.45%) and strong augmentation (60.47%).

3. **Hyperparameter sensitivity is low in the optimal region.** GridSearchCV over PCA components [20, 25, 30, 35, 40] and SVM C [3, 4, 5, 6, 8] shows a broad plateau around PCA 35–40 and $C = 4$ (CV 95.35%), confirming robust parameter choices.

4. PCA dimensionality reduction significantly benefits non-linear classifiers by removing noise and reducing the curse of dimensionality. Optimal component count is 35–50 (explaining 80–83% variance).

5. Non-linear methods substantially outperform linear baselines, confirming that digit classes require flexible decision boundaries.

6. Only 1.5% of samples are universally misclassified (hard samples), while 14.1% represent model-capability boundaries where better models can improve.

7. The domain shift between MNIST and challenge data is substantial for non-augmented models (20–25 pp drop), but data augmentation reduces this gap to 13.2 pp, demonstrating its effectiveness for learning writer-invariant features.

Contribution Statement

Both group members contributed equally to this project. Yikai LAI was responsible for implementing the experiment pipeline, classifier wrappers, and the unified result management system. Zhang Yao contributed to the experimental design, error analysis scripts, and challenge evaluation. Both members participated in writing the project report and analyzing the results.

References

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